

Theorem C is a corollary of Rick's Theorem 2.5,  
and the derivation is a telescoping with one broken step

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|                   |                                                           |
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| <b>Repository</b> | <code>clio-vega/proofs</code>                             |
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*Provenance block added 2026-10-09 at Rick's request (email UID 790, 2026-10-08 08:42). Per PROTOCOL §2.3 a PDF with no commit hash cannot be checked against the record; this one now can.*

### Abstract

I close `gap:rick` of `2026-10-07-two-part-green-polynomials.tex`. Rick's Theorem 2.5 (the *Two-point formula*, labelled `thm:2pt` in his FPSAC 2027 draft) is a closed form for  $\langle T_{ag}, p_x p_y \rangle$  for *every* symmetric  $g$  in  $a$  variables; transported by his dictionary it therefore computes  $Y_{(x,y)}^\lambda(t)$  for *every*  $\lambda$ . My Theorem C is its specialisation to  $\ell(\lambda) = 2$ , and I prove the specialisation: his formula, read at  $a = 2$ , is a linear functional on the Laurent coefficients of  $G_1(w) = P_\rho(1, w; t)$  which *telescopes* against the potential  $\alpha(k) = (t^{-k} + t^{k-1})/(1+t)$  at every index except  $e = 0$ , and the single broken step is exactly the source of the integer terms in Theorem C. So the brief's outcome 2 holds:  $A = B$ , **after a rewrite that is not a substitution**. The rewrite is verified symbolically in  $\mathbb{Q}(t)$  on all 271 ordered rows with  $|\lambda| \leq 12$ , and against two ground truths — one of which uses no tableaux at all — with 0 disagreements. Consequences I volunteer: Theorem C is **demoted to a corollary** with an independent proof; Theorem B is *not* a corollary, because it is a statement about the  $h$ -expansion rather than about values; and Theorem B composed with Theorem 2.5 gives a closed form for  $[h_\nu]Q'_\lambda$  for *all*  $\lambda$  and all  $\ell(\nu) \leq 2$ , which is the item my own paper listed as not attempted.

## 1 Conventions

All conventions are those of `2026-10-07-two-part-green-polynomials.tex`: the ordinary Hall pairing  $\langle \cdot, \cdot \rangle$  with  $\langle p_\lambda, p_\mu \rangle = \delta_{\lambda\mu} z_\lambda$ ; the Hall–Littlewood pairing  $\langle p_\lambda, p_\mu \rangle_t = \delta_{\lambda\mu} z_\lambda \prod_i (1 - t^{\lambda_i})^{-1}$ ;  $\varphi_m(t) = \prod_{k=1}^m (1 - t^k)$ ;  $b_\mu(t) = \prod_{i \geq 1} \varphi_{m_i(\mu)}(t)$ ;  $n(\mu) = \sum_i (i-1)\mu_i$ ;  $Q'_\mu = \sum_\nu K_{\nu\mu}(t) s_\nu$  with  $K$  in the charge normalisation; and

$$Y_\rho^\lambda(t) = \langle p_\rho, Q'_\lambda \rangle = [P_\lambda] p_\rho = \sum_\nu K_{\nu\lambda}(t) \chi_\rho^\nu. \quad (1)$$

Rick's  $X_\mu^\lambda$  of  $p_\mu = \sum_\lambda X_\mu^\lambda(t) P_\lambda$  is literally  $Y_\mu^\lambda$ ; this needs no computation, since that equation pins  $X$  with no normalisation freedom. Throughout,  $\lambda = (a, b)$  with  $a \geq b \geq 1$ ,  $n = a + b$ ,  $\rho = (a-1, b-1)$ ,  $m = a - b$ , and  $\mu = \{x, y\}$  is a two-part class of  $n$ . I write  $\llbracket \cdot \rrbracket$  for an indicator.

**Theorem 1** (C, restated from the earlier paper). *With  $m_0 = \min(x, y)$ ,*

$$Y_{\{x,y\}}^{(a,b)}(t) = \begin{cases} 1 + \llbracket a = b \rrbracket + (t-1)t^{b-1}, & m_0 = b, \\ (t-1)t^{b-1} + (t-1)t^{b-m_0-1}, & 1 \leq m_0 \leq b-1, \\ (t-1)t^{b-1}, & m_0 > b. \end{cases}$$

**Remark 2** (this corner is classical). *Added 2026-10-09 at Rick’s request (email UID 789, 2026-10-08 08:39), so that a referee does not have to raise it.* The restriction of Theorem 1 to a two-row shape  $\lambda = (a, b)$  paired against a two-part class is not new, and it is not deep. It follows in three classical steps: the Kostka–Foulkes polynomial for a two-row shape and two-part content is the monomial  $K_{\nu, (a,b)}(t) = t^{b-\nu_2}$  (one tableau; Macdonald III (6.5)(ii) supplies monicity, and the tableau count is Young’s rule), whence the  $h$ -expansion of  $Q'_\lambda$  collapses to a two-term pairing. Rick reached the same closed form independently by exactly this route (`grandpa-rick/rick-research, notes/for-clio/2026-10-08-two-row-green-is-young.tex`; his FPSAC Example 6.5), and the formula also follows from the Jing–Liu recurrence (`arXiv:2104.04411`, §2), which predates both of us and to which priority for the two-row Green formula belongs.

What is *not* classical, and what this paper and its predecessor are actually about, is the passage off this corner: the length filtration of  $M_{\rho\nu} = \langle p_\rho, h_\nu \rangle$  and the behaviour at general  $\ell(\lambda)$ , and above all Theorem D, the *nonexistence* of a product form. Theorem D is untouched by this remark — a closed form and a product form are different predicates, and the separator is product-versus-sum alone (see `scope-reconciliation-corrects-my-record` in the registry). A reader should take Theorem 1 as a convenient normalisation, correctly credited above, and not as a claim of novelty on the two-row slice.

## 2 gap:rick closed: the statement was on disk under a different label

gap:rick read: “the statement itself is in `grandpa-rick/work-in-progress` at 71b4cad, which I do not hold.” I hold it, and have since 2026-10-06. It is at

`/tmp/rick-review/fpsac2027/fpsac2027-draft.tex`, environment `thm:2pt`, *Two-point formula*.

The string 2.5 does not occur anywhere near it. The identification is made by the *registry comment line immediately above the environment*,

```
% registry: two-point-string-formula-thm25 (+ shuffle-identity-lemma24), grade:
               proved
```

together with `/tmp/rick-review/registry/hikita-star-dominance-support.json`, whose node `two-point-string-formula-thm25` carries approach: “Thm 2.5: closed  $\langle T_{ag, p_x p_y} \rangle$  for all symmetric  $g$ .” Three further corroborations, each independent of that one:

- (i) the same node’s `novelty_wake226` field states the dictionary  $\Phi_a(P_\rho; x, y) = (1 - t^x)(1 - t^y)X_{(x,y)}^\lambda / b_\lambda$  with  $\lambda = \rho + 1^a$ , which is the dictionary his 2026-10-07 note attributes to Theorem 2.5;
- (ii) his §3(b) substitution in that note is reproduced *term for term* by setting  $a = 2$  in `thm:2pt` (§3.1 below);

- (iii) his prose description of Theorem 2.5 — “explicit, with no sum over classes, linear in the two-string specialisation  $G_A$  of  $P_\rho$ ; not a product formula” — is the shape of `thm:2pt` and of nothing else in the draft.

**Remark 3** (the failure mode, named). This is a third distinct way a holdings record can be wrong, and the sharpest so far. The two I had recorded were *a filename is a claim about contents* and *a scratch directory is invisible to an audit that only scans designated corpora*. This one is: **a theorem number is a claim about a numbering, not about contents**. I searched for “Theorem 2.5” in a document that numbers its theorems by L<sup>A</sup>T<sub>E</sub>X label and section, and concluded absence. The instrument that resolved it was his own `% registry:` comment — a channel I had read past for two sessions, in the same file, four lines above the thing I said I did not hold.

### 3 Rick’s Theorem 2.5, verbatim

From his Prop. 2.1 (subset formula) notation: for  $F \in \Lambda_N$ ,  $c_A = \prod_{i \in A, j \notin A} \frac{x_i - tx_j}{x_i - x_j}$  and  $X_A = \prod_{i \in A} x_i$ ; and  $T_ag := \sum_{|A|=a} c_A X_A g(X_A)$  for  $g \in \Lambda_a$ . His Notation paragraph declares  $\langle \cdot, \cdot \rangle$  to be *the Hall pairing* and  $\langle \cdot, \cdot \rangle_t$  the HL pairing, so the bracket in the statement below is unambiguous.

**Theorem 4** (Rick, Thm. 2.5 = `thm:2pt`; quoted). *Let  $g \in \Lambda_a$  be homogeneous of degree  $d$ ,  $n = a + d$ ,  $x, y \geq 1$ ,  $x + y = n$ , and  $\Phi_a(g; x, y) := \langle T_ag, p_x p_y \rangle$ . For  $1 \leq A \leq a - 1$  put  $B = a - A$  and  $G_A(w) = \sum_k G_{A,k} w^k := g(1, t, \dots, t^{A-1}, w, wt, \dots, wt^{B-1})$ . Then*

$$\Phi_a(g; x, y) = (-1)^a (1-t^x)(1-t^y) \left\{ \sum_{A=1}^{a-1} t^{-AB} \left[ \frac{G_{A,y-B}}{(1-t^A)(1-t^B)} + \frac{1}{1-t^a} \sum_{m \geq 1} (t^{-Am} - t^{Bm}) G_{A,y-B-m} \right] - \frac{g(1, t, \dots, t^{a-1})}{1-t^a} \sum_{j=0}^{a-1} t^{-jy} \right\}.$$

**Lemma 5** (his dictionary = my `lem:dict`). *For  $\lambda = \rho + 1^a$  with  $a = \ell(\lambda)$ ,  $\Phi_a(P_\rho; x, y) = (1-t^x)(1-t^y) \frac{Y_{(x,y)}^\lambda(t)}{b_\lambda(t)}$ .*

Two things about the quantifiers in Theorem 4, both of which I checked by reading the statement rather than inferring from its use. First,  $g$  is an *arbitrary* homogeneous element of  $\Lambda_a$ . Second,  $\lambda = \rho + 1^a$  with  $a = \ell(\lambda)$  places *no* restriction on  $\lambda$ : every partition is  $\rho + 1^{\ell(\lambda)}$  for  $\rho = \lambda - 1^{\ell(\lambda)}$ . Hence:

**Verdict 6** (Task 3, first half). Theorem 4 together with Lemma 5 is a closed form for the *entire* two-part-class slice  $\{Y_{(x,y)}^\lambda : \lambda \vdash n, x + y = n\}$ , for all  $\lambda$ , not only two-row  $\lambda$ . The slice of my Theorem B is *inside* his, and the slice of my Theorem C is inside that.

Verdict 6 is a statement about the hypotheses, but I also tested it: his closed form, coded verbatim for general  $a$ , agrees with ground truth on all  $\lambda \vdash n$  for  $n \leq 8$  —  $\ell(\lambda) = 1, \dots, 8$  — in 2100 comparisons, 0 disagreements, broken down by length in §6.

### 3.1 His $a = 2$ substitution is a correct specialisation of his own theorem

Set  $a = 2$  in Theorem 4. Then  $A = B = 1$  is the only term;  $t^{-AB} = t^{-1}$ ;  $(1-t^A)(1-t^B) = (1-t)^2$ ;  $1-t^a = 1-t^2$ ;  $\sum_{j=0}^1 t^{-jy} = 1+t^{-y}$ ;  $(-1)^a = +1$ ; and  $G_1(w) = g(1, w)$ . Every one of these matches his §3(b) expression character for character. Outcome 3 of the brief (“his substitution is wrong”) is therefore excluded *by inspection*, before any numerical work. I nevertheless carry his §3(b) expression as a separate column (B') throughout, transcribed from his note and not re-derived, so that the claim is tested and not merely asserted.

For  $g = P_\rho$  with  $\rho = (a-1, b-1)$  in two variables,

$$G_1(w) = P_\rho(1, w; t) = w^{b-1} \left( 1 + w^m + (1-t) \sum_{0 < k < m} w^k \right), \quad G_1 = w^{b-1} \text{ if } m = 0, \quad (2)$$

so the exponent support of  $G_1$  is the integer interval  $[b-1, a-1]$  with coefficient 1 at the two endpoints and  $1-t$  strictly inside; and

$$G_1(t) = P_\rho(1, t; t) = \begin{cases} t^{b-1}(1+t), & a > b, \\ t^{b-1}, & a = b, \end{cases} \quad (3)$$

because for  $m \geq 1$ ,  $1+t^m + (1-t)(t + \dots + t^{m-1}) = 1+t^m + (t-t^m) = 1+t$ . (Both (2) and (3) are the  $n = 2$  case of Macdonald's principal specialisation, but neither needs it; I do not hold the book and quote no equation number from it.)

## 4 The specialisation, proved

### 4.1 A linear functional

For a Laurent polynomial  $F(w) = \sum_k F_k w^k$  put

$$L_y(F) := t^{-1} \left[ \frac{F_{y-1}}{(1-t)^2} + \frac{1}{1-t^2} \sum_{j \geq 1} (t^{-j} - t^j) F_{y-1-j} \right] - \frac{F(t)(1+t^{-y})}{1-t^2}, \quad (4)$$

so that, by §3.1 and Lemma 5,

$$Y_{(x,y)}^{(a,b)}(t) = b_{(a,b)}(t) \cdot L_y(G_1). \quad (5)$$

$L_y$  is linear, so evaluate it on a monomial. Put  $e := y-1-c$ . Then  $F_{y-1} = \llbracket e = 0 \rrbracket$  and the  $j$ -sum retains only  $j = e$ , and only when  $e \geq 1$ :

$$L_y(w^c) = \Psi(e) - \frac{t^c(1+t^{-y})}{1-t^2}, \quad \Psi(e) := \frac{t^{-1} \llbracket e = 0 \rrbracket}{(1-t)^2} + \frac{t^{-1}(t^{-e} - t^e) \llbracket e \geq 1 \rrbracket}{1-t^2}, \quad (6)$$

with  $\Psi(e) = 0$  for  $e \leq -1$ . **Everything turns on the sign of  $e$ :**  $e \geq 0$  iff  $c \leq y-1$ , so his formula sees only the part of  $G_1$  at or below  $w^{y-1}$ . That one-sided truncation is the whole of the apparent  $x \leftrightarrow y$  asymmetry of Theorem 4.

## 4.2 The potential

Put  $\psi(e) := (1-t)^2 \Psi(e)$ , so  $\psi(e) = 0$  for  $e < 0$ ,  $\psi(0) = t^{-1}$ , and

$$\psi(e) = \frac{(1-t)(t^{-e} - t^e)}{t(1+t)} = (1-t)^2 t^{-1-e} (1+t^2 + \dots + t^{2e-2}) \quad (e \geq 1).$$

Define

$$\boxed{\alpha(k) := \frac{t^{-k} + t^{k-1}}{1+t}} \quad (k \in \mathbb{Z}). \quad (7)$$

$\alpha(k)$  is a Laurent polynomial: for  $k \geq 1$  the numerator is  $t^{-k}(1+t^{2k-1})$  and  $2k-1$  is odd, so  $1+t$  divides it; for  $k \leq 0$  likewise. Three identities, each one line:

$$(T1) \quad \psi(e) = \alpha(e+1) - \alpha(e) \quad (e \geq 1), \quad (8)$$

$$(T2) \quad t\alpha(k+1) - \alpha(k) = (t-1)t^{k-1}, \quad \alpha(k+1) - t\alpha(k) = (1-t)t^{-k-1} \quad (k \in \mathbb{Z}), \quad (9)$$

$$(T3) \quad \alpha(0) = \alpha(1) = t^{-1}. \quad (10)$$

*Proof.* (T1): both sides equal  $(1-t)^2 t^{-1-e} (1-t^{2e}) / (1-t^2)$ , since  $\alpha(e+1) - \alpha(e) = \frac{t^{-e-1} + t^e - t^{-e} - t^{e-1}}{1+t} = \frac{t^{-e-1}(1-t)(1-t^{2e+1}/t^2)}{1+t}$  — more directly, clearing  $1+t$ ,  $(1+t)(\alpha(e+1) - \alpha(e)) = t^{-e-1} - t^{-e} + t^e - t^{e-1} = -t^{-e-1}(t-1) - t^{e-1}(1-t) = (1-t)(t^{-e-1} - t^{e-1})$ , and  $(1+t)\psi(e) = (1-t)t^{-1}(t^{-e} - t^e) = (1-t)(t^{-e-1} - t^{e-1})$ . (T2):  $(1+t)(t\alpha(k+1) - \alpha(k)) = t^{-k} + t^{k+1} - t^{-k} - t^{k-1} = t^{k-1}(t^2 - 1)$ , and dividing by  $1+t$  gives  $-t^{k-1}(1-t)$ ; similarly  $(1+t)(\alpha(k+1) - t\alpha(k)) = t^{-k-1} + t^k - t^{-k-1} - t^k = t^{-k-1}(1-t^2)$ . (T3):  $\alpha(0) = (1+t^{-1})/(1+t) = t^{-1}$  and  $\alpha(1) = (t^{-1} + 1)/(1+t) = t^{-1}$ .  $\square$

**Remark 7** (the broken step, which is the content). By (T3) the telescoped increment at  $e = 0$  is  $\alpha(1) - \alpha(0) = 0$ , whereas  $\psi(0) = t^{-1} \neq 0$ . So  $\psi$  **telescopes against  $\alpha$  at every index  $e \geq 1$  and fails to telescope at exactly one index,  $e = 0$** ; it also fails for  $e \leq -1$ , where  $\psi = 0$  but  $\alpha(0) - \alpha(-1) = -t^{-2}(1-t)^2 \neq 0$ . The lone surviving  $t^{-1}$  at  $e = 0$  is what produces the integer terms  $1, 1 + \llbracket a = b \rrbracket, 2$  of Theorem 1; every other term of Theorem 1 is a telescoping boundary value. This is the whole mechanism, and it is ablated in §7: replacing  $\psi$  by the naive telescoped increment at all  $e$  drops agreement from 110/110 to 20/110.

## 4.3 Assembly

Put  $E_1 := y - a$  and  $E_2 := y - b$ , so  $E_2 - E_1 = m$ ,  $-E_2 - 1 = b - y - 1$ , and  $-E_1 = x - b$ . Since  $x, y \geq 1$  we have  $E_1 \leq b - 1$  and  $E_2 \geq 1 - b$ .

**Case  $a > b$ .** Here  $b_\lambda = \varphi_1^2 = (1-t)^2$  and  $(1-t)^2/(1-t^2) = (1-t)/(1+t)$ , so by (3) the subtracted term of (4), scaled by  $b_\lambda$ , is exactly  $-(1-t)t^{b-1}(1+t^{-y})$ . Summing (6) over the monomials of  $G_1$  — endpoints  $c = b-1, a-1$ , i.e.  $e = E_2, E_1$ ; interior  $c \in [b, a-2]$ , i.e.  $E_1 < e < E_2$  — gives

$$Y_{(x,y)}^{(a,b)} = \psi(E_1) + \psi(E_2) + (1-t) \sum_{E_1 < e < E_2} \psi(e) - (1-t)t^{b-1}(1+t^{-y}). \quad (\star)$$

Note that  $(\star)$  reproduces the shape of  $G_1$  itself: endpoints weight 1, interior weight  $1-t$ . Five subcases exhaust the signs, and  $E_1 = E_2 = 0$  cannot occur because  $m \geq 1$ .

- (i)  $E_1 \geq 1$  (so  $E_2 \geq 2$ ). Then  $y \geq a + 1$ , hence  $x \leq b - 1$  and  $m_0 = x$ . All indices are  $\geq 1$ , so  $\sum_{E_1 < e < E_2} \psi = \alpha(E_2) - \alpha(E_1 + 1)$  by (T1) — correct also when  $m = 1$ , where the sum is empty and  $E_2 = E_1 + 1$ . With (T1) the brace of  $(\star)$  becomes

$$[t\alpha(E_1+1) - \alpha(E_1)] + [\alpha(E_2+1) - t\alpha(E_2)] \stackrel{(T2)}{=} (t-1)t^{E_1-1} + (1-t)t^{-E_2-1}.$$

The second term is  $(1-t)t^{b-y-1}$  and **cancels** against  $-(1-t)t^{b-1-y}$  in  $(\star)$ . With  $E_1 - 1 = b - x - 1$  this leaves  $(t-1)t^{b-1} + (t-1)t^{b-m_0-1}$ : Theorem 1, branch  $1 \leq m_0 \leq b - 1$ .

- (ii)  $E_1 \leq -1$ ,  $E_2 \geq 1$ . Then  $y \geq b + 1$  and  $x \geq b + 1$ , so  $m_0 > b$ .  $\psi(E_1) = 0$ ; the interior contributes  $\psi(0) = t^{-1}$  plus  $\alpha(E_2) - \alpha(1)$ , and by (T3) the  $t^{-1}$  **cancels**  $\alpha(1)$ , leaving  $\alpha(E_2)$ . The brace is  $\alpha(E_2+1) - t\alpha(E_2) = (1-t)t^{b-y-1}$ , which cancels as before. Hence  $Y = (t-1)t^{b-1}$ : branch  $m_0 > b$ .
- (iii)  $E_2 = 0$ , i.e.  $y = b$ ,  $x = a$ ,  $m_0 = b$ . Only  $\psi(0) = t^{-1}$  survives, and  $-(1-t)t^{b-1}t^{-b} = -(1-t)t^{-1} = -t^{-1} + 1$  absorbs it. Hence  $Y = 1 + (t-1)t^{b-1}$ : branch  $m_0 = b$  with  $\llbracket a = b \rrbracket = 0$ .
- (iv)  $E_2 \leq -1$ , i.e.  $y \leq b - 1$ ,  $m_0 = y$ . Every  $\psi$  vanishes and  $Y = (t-1)t^{b-1} + (t-1)t^{b-y-1}$ : branch  $1 \leq m_0 \leq b - 1$ .
- (v)  $E_1 = 0$ , i.e.  $y = a$ ,  $x = b$ ,  $m_0 = b$ ,  $E_2 = m \geq 1$ . The brace is  $t^{-1} + \alpha(E_2+1) - t\alpha(E_2) - (1-t)\alpha(1)$ , and (T3) turns the last term into  $-(1-t)t^{-1} = -t^{-1} + 1$ ; the remaining  $(1-t)t^{b-y-1}$  cancels. Hence  $Y = 1 + (t-1)t^{b-1}$ : branch  $m_0 = b$ .

**Case  $a = b$ .** Now  $G_1 = w^{b-1}$  alone,  $b_\lambda = \varphi_2 = (1-t)(1-t^2)$ , so  $b_\lambda \Psi(e) = (1+t)\psi(e)$ ,  $G_1(t) = t^{b-1}$ , and with  $E := y - b$ ,

$$Y_{(x,y)}^{(b,b)} = (1+t)\psi(E) - (1-t)t^{b-1}(1+t^{-y}).$$

Three subcases, and  $m_0 > b$  is impossible because  $m_0 \leq n/2 = b$ :

- $E = 0$  ( $x = y = b$ ):  $(1+t)t^{-1} = t^{-1} + 1$  and  $-(1-t)t^{-1} = -t^{-1} + 1$ , so  $Y = 2 + (t-1)t^{b-1}$ . This is the *only* place the  $\llbracket a = b \rrbracket$  term of Theorem 1 is exercised, and it arises from  $b_\lambda = \varphi_2$  rather than  $\varphi_1^2$  — i.e. from the dictionary, not from the formula.
- $E \leq -1$  ( $y < b$ ,  $m_0 = y$ ):  $\psi = 0$  and  $Y = (t-1)t^{b-1} + (t-1)t^{b-y-1}$ .
- $E \geq 1$  ( $y > b$ ,  $m_0 = x = 2b - y$ ):  $(1+t)\alpha(k) = t^{-k} + t^{k-1}$ , so  $(1+t)\psi(E) = t^{b-y-1} + t^{y-b} - t^{b-y} - t^{y-b-1}$ ; the terms  $t^{b-y-1}$  and  $t^{b-y}$  cancel against  $-(1-t)t^{b-1-y}$ , leaving  $Y = (t-1)t^{b-1} + (t-1)t^{y-b-1}$  with  $y - b - 1 = b - m_0 - 1$ .

**Theorem 8** (the specialisation). *For every  $a \geq b \geq 1$  and every  $x, y \geq 1$  with  $x + y = a + b$ , Rick's Theorem 4 at  $a = 2$  with  $g = P_{(a-1, b-1)}$ , transported by Lemma 5, equals the right-hand side of Theorem 1, as an identity in  $\mathbb{Q}(t)$ . Consequently Theorem C is a corollary of Theorem 2.5, and — since the derivation above is reversible, being a chain of identities — the independent proof of Theorem C in the earlier paper is a proof of Theorem 2.5 restricted to  $a = 2$ ,  $g = P_\rho$ .*

**Verdict 9** (Task 2). **Outcome 2 of the brief:**  $A = B$ , after a rewrite that is not a substitution. The transition map is exhibited:  $L_y$  of (4) read monomial-wise on  $G_1$ , telescoped against  $\alpha$  of (7), with the one broken index of Remark 7.

**Remark 10** (what the rewrite actually buys, stated without inflation). Theorem 1 has three features that Theorem 4 does not display and that the telescoping is what makes visible.

- (1) *Manifest  $x \leftrightarrow y$  symmetry.*  $\Phi_a(g; x, y)$  is symmetric by definition, but the closed form of Theorem 4 is written one-sidedly in  $y$ ; at  $a = 2$  the symmetry holds only after the cancellations above. (Checked: both orderings of every  $\mu$  are separate rows of the table.)
- (2) *Manifest Laurent-polynomiality.* Theorem 4 at  $a = 2$  is a sum of terms with poles at  $t = 0, \pm 1$ ; the output has none.
- (3) *Independence of  $a$  except through  $\llbracket a = b \rrbracket$ .* The  $a$ -dependence of Theorem 4's input sits in the upper endpoint  $a - 1$  of  $G_1$ 's support and in (3); the telescoping is precisely what makes the upper-endpoint contribution cancel.

These are properties of a normal form, not new values. I claim no more for them than that.

## 5 The table

Five columns were computed, each from a different mechanism:

| col | what                                                                           | mechanism                                                                                                                 |
|-----|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| A   | Theorem 1, closed form                                                         | charge $\rightarrow$ <code>lem:mono</code> $\rightarrow$ <code>prop:tworow</code> $\rightarrow$ Thm B $\rightarrow$ Thm C |
| A'  | Theorem 1, indicator form                                                      | a second transcription of the same statement                                                                              |
| B   | Theorem 4, coded verbatim for general $a$                                      | his $t$ -strings + Lemma 5                                                                                                |
| B'  | his §3(b) $a = 2$ substitution, transcribed                                    | his substitution + Lemma 5                                                                                                |
| C1  | $\sum_{\nu} K_{\nu\lambda}(t) \chi_{\rho}^{\nu}$                               | charge Kostka–Foulkes $\times$ Murnaghan–Nakayama                                                                         |
| C2  | $b_{\lambda} \langle p_{\rho}, P_{\lambda} \rangle / \prod_i (1 - t^{\rho_i})$ | Gram–Schmidt in the $p$ -basis + plethysm. <b>No tableaux.</b>                                                            |

A versus C2 is the only pair that shares no machinery: A reaches the value through charge and the  $h$ -expansion, C2 through orthogonalisation for  $\langle \cdot, \cdot \rangle_t$  and the plethystic definition of  $Q'$ . A versus C1 shares the charge statistic, so it is a weaker control and is reported separately. B versus C2 shares Lemma 5 with B; §6 decouples that.

**There were no disagreements, at any row, in any pair, at any value of  $t$ .** Since there is no witness to report, what follows is the common value, and then — in §6–§7 — the counts, the stratum split, and the rank line, which are the parts of a green table that carry information.

| $\lambda$ | $\mu$  | branch | stratum                  | common value of the five columns |
|-----------|--------|--------|--------------------------|----------------------------------|
| $n = 2$   |        |        |                          |                                  |
| (1, 1)    | (1, 1) | $m=b$  | hook, $a=b, x=y, \min=1$ | $t + 1$                          |
| $n = 3$   |        |        |                          |                                  |
| (2, 1)    | (2, 1) | $m=b$  | hook, $\min=1$           | $t$                              |
| $n = 4$   |        |        |                          |                                  |
| (3, 1)    | (3, 1) | $m=b$  | hook, $\min=1$           | $t$                              |
| (3, 1)    | (2, 2) | $m>b$  | hook, $x=y$              | $t - 1$                          |
| (2, 2)    | (3, 1) | $m<b$  | $a=b, \min=1$            | $(t - 1)(t + 1)$                 |
| (2, 2)    | (2, 2) | $m=b$  | $a=b, x=y$               | $t^2 - t + 2$                    |
| $n = 5$   |        |        |                          |                                  |
| (4, 1)    | (4, 1) | $m=b$  | hook, $\min=1$           | $t$                              |
| (4, 1)    | (3, 2) | $m>b$  | hook                     | $t - 1$                          |
| (3, 2)    | (4, 1) | $m<b$  | $\min=1$                 | $(t - 1)(t + 1)$                 |
| (3, 2)    | (3, 2) | $m=b$  | <i>generic</i>           | $t^2 - t + 1$                    |

| $\lambda$ | $\mu$  | branch | stratum        | common value                  |
|-----------|--------|--------|----------------|-------------------------------|
| $n = 6$   |        |        |                |                               |
| (5, 1)    | (5, 1) | $m=b$  | hook, min =1   | $t$                           |
| (5, 1)    | (4, 2) | $m>b$  | hook           | $t - 1$                       |
| (5, 1)    | (3, 3) | $m>b$  | hook, $x=y$    | $t - 1$                       |
| (4, 2)    | (5, 1) | $m<b$  | min =1         | $(t - 1)(t + 1)$              |
| (4, 2)    | (4, 2) | $m=b$  | <i>generic</i> | $t^2 - t + 1$                 |
| (4, 2)    | (3, 3) | $m>b$  | $x=y$          | $t(t - 1)$                    |
| (3, 3)    | (5, 1) | $m<b$  | $a=b$ , min =1 | $t(t - 1)(t + 1)$             |
| (3, 3)    | (4, 2) | $m<b$  | $a=b$          | $(t - 1)(t^2 + 1)$            |
| (3, 3)    | (3, 3) | $m=b$  | $a=b$ , $x=y$  | $(t + 1)(t^2 - 2t + 2)$       |
| $n = 7$   |        |        |                |                               |
| (6, 1)    | (6, 1) | $m=b$  | hook, min =1   | $t$                           |
| (6, 1)    | (5, 2) | $m>b$  | hook           | $t - 1$                       |
| (6, 1)    | (4, 3) | $m>b$  | hook           | $t - 1$                       |
| (5, 2)    | (6, 1) | $m<b$  | min =1         | $(t - 1)(t + 1)$              |
| (5, 2)    | (5, 2) | $m=b$  | <i>generic</i> | $t^2 - t + 1$                 |
| (5, 2)    | (4, 3) | $m>b$  | <i>generic</i> | $t(t - 1)$                    |
| (4, 3)    | (6, 1) | $m<b$  | min =1         | $t(t - 1)(t + 1)$             |
| (4, 3)    | (5, 2) | $m<b$  | <i>generic</i> | $(t - 1)(t^2 + 1)$            |
| (4, 3)    | (4, 3) | $m=b$  | <i>generic</i> | $t^3 - t^2 + 1$               |
| $n = 8$   |        |        |                |                               |
| (7, 1)    | (7, 1) | $m=b$  | hook, min =1   | $t$                           |
| (7, 1)    | (6, 2) | $m>b$  | hook           | $t - 1$                       |
| (7, 1)    | (5, 3) | $m>b$  | hook           | $t - 1$                       |
| (7, 1)    | (4, 4) | $m>b$  | hook, $x=y$    | $t - 1$                       |
| (6, 2)    | (7, 1) | $m<b$  | min =1         | $(t - 1)(t + 1)$              |
| (6, 2)    | (6, 2) | $m=b$  | <i>generic</i> | $t^2 - t + 1$                 |
| (6, 2)    | (5, 3) | $m>b$  | <i>generic</i> | $t(t - 1)$                    |
| (6, 2)    | (4, 4) | $m>b$  | $x=y$          | $t(t - 1)$                    |
| (5, 3)    | (7, 1) | $m<b$  | min =1         | $t(t - 1)(t + 1)$             |
| (5, 3)    | (6, 2) | $m<b$  | <i>generic</i> | $(t - 1)(t^2 + 1)$            |
| (5, 3)    | (5, 3) | $m=b$  | <i>generic</i> | $t^3 - t^2 + 1$               |
| (5, 3)    | (4, 4) | $m>b$  | $x=y$          | $t^2(t - 1)$                  |
| (4, 4)    | (7, 1) | $m<b$  | $a=b$ , min =1 | $t^2(t - 1)(t + 1)$           |
| (4, 4)    | (6, 2) | $m<b$  | $a=b$          | $t(t - 1)(t^2 + 1)$           |
| (4, 4)    | (5, 3) | $m<b$  | $a=b$          | $(t - 1)(t + 1)(t^2 - t + 1)$ |
| (4, 4)    | (4, 4) | $m=b$  | $a=b$ , $x=y$  | $t^4 - t^3 + 2$               |

## 6 Counts, strata, and the rank line

**What was compared.** At  $|\lambda| \leq 8$  there are 44 distinct pairs  $(\lambda, \mu)$  with  $\mu$  an unordered two-part class, and 78 *ordered* evaluations  $(x, y)$  — each unordered  $\mu$  with  $x \neq y$  is evaluated twice, which is a real test because column B is written asymmetrically in  $y$ . At  $|\lambda| \leq 12$  these become 146 and 271.

- **Symbolic in  $\mathbb{Q}(t)$ ,  $|\lambda| \leq 7$  (50 ordered rows):** all of A, A', B, B', C1, C2 pairwise — 0 disagreements. (`table7.log`)
- **Symbolic in  $\mathbb{Q}(t)$ ,  $|\lambda| \leq 12$  (271 ordered rows):**  $B \equiv B' \equiv (\star) \equiv A$  — 0 disagreements. This is the full algebraic spine of §4 and needs no ground truth. (`spine.log`)
- **Numerically exact,  $|\lambda| \leq 12$ , 40 exact rational  $t$**  (including 10 negative values and values of large height): all five columns, 65,040 comparisons — 0 disagreements. (`sweep.log`) Gram–Schmidt was redone at each  $t$  in rational arithmetic; no floating point anywhere in this document.
- **Theorem 4 on the full slice, all  $\lambda \vdash n$ ,  $n \leq 8$ :** B versus C2, 2100 comparisons, 0 disagreements, distributed over  $\ell(\lambda)$  as {1:168, 2:468, 3:522, 4:420, 5:252, 6:150, 7:78, 8:42}. This is the evidence behind Verdict 6: his theorem is right off the two-row locus too.

**Decoupling Theorem 4 from Lemma 5.** Everything above tests the *composite* (closed form)◦(dictionary), which cannot tell a compensating pair of errors from two correct statements. I therefore implemented  $T_{ag} = \sum_{|A|=a} c_A X_A g(X_A)$  from his Prop. 2.1 directly, and computed  $\Phi_a(g; x, y) = \langle T_{ag}, p_x p_y \rangle$  from the definition via  $\langle F, p_x p_y \rangle = \sum_{\nu} d_{\nu} [m_{\nu}] F$  where  $p_x p_y = \sum_{\nu} d_{\nu} h_{\nu}$  (legitimate because  $\langle m_{\kappa}, h_{\nu} \rangle = \delta_{\kappa\nu}$ ), taking  $N = n$  variables, which suffices since a symmetric function of degree  $n$  has  $m_{\kappa}$ -support on  $\kappa \vdash n$  only. Against that definition, *separately*: his closed form, 0 disagreements; and his dictionary, 0 disagreements. **The reach is  $|\lambda| \leq 6$ : 32 ordered rows** — it is a symbolic computation of  $\binom{n}{2}$  rational-function terms in  $n$  variables, and  $n = 7$  did not finish in the session (`decouple.log`). Off that range I have the composite only, recorded as Gap 20. A by-product: the agreement of my  $N = n$  computation with his  $N$ -independent closed form corroborates the stability in  $N$  that his Prop. 2.1 asserts.

**Degenerate versus generic.** Rows carrying any of:  $b = 1$  ( $\lambda$  a hook),  $a = b$  ( $m = 0$ , where  $G_1$  collapses to a single monomial),  $x = y$ ,  $\min(x, y) = 1$ .

|                                                   | $ \lambda  \leq 8$ | $ \lambda  \leq 12$ |
|---------------------------------------------------|--------------------|---------------------|
| ordered rows                                      | 78                 | 271                 |
| branch $m_0 = b$ / $m_0 < b$ / $m_0 > b$          | 28 / 28 / 22       | 66 / 110 / 95       |
| hook $b = 1$                                      | 28                 | 66                  |
| square $a = b$                                    | 16                 | 36                  |
| $x = y$                                           | 10                 | 21                  |
| $\min(x, y) = 1$                                  | 31                 | 71                  |
| <b>generic (no tag)</b>                           | <b>20</b>          | <b>120</b>          |
| branch split among generic                        | 12 / 4 / 4         | 40 / 40 / 40        |
| rows with $b \geq 2$ , $a > b$ , $1 \leq m_0 < b$ | 16                 | 80                  |

So at  $|\lambda| \leq 8$  **only 20 of 78 rows are generic**, and the brief’s warning is justified: a table read as “78 checks” is in fact 20 generic checks with 58 rows on one or more degenerate strata. At  $|\lambda| \leq 12$  the generic fraction improves to 120/271 and all three branches are populated 40/40/40.

**The rank line.** The honest rank statistic here is not the dimension of a span of linear relations — each row pins one unknown  $Y_{\mu}^{\lambda}$  by five mechanisms, so the system is diagonal — but *how many distinct values the rows take*:

|                                              | $ \lambda  \leq 8$ | $ \lambda  \leq 12$ |
|----------------------------------------------|--------------------|---------------------|
| distinct values of $Y$ over the ordered rows | 16 of 78           | 31 of 271           |
| values realised with $\geq 2$ distinct $a$   | 8 of 16            | 19 of 31            |
| values realised with $\geq 2$ distinct $b$   | 0 of 16            | 0 of 31             |
| non-monomial values                          | 66 of 78           | 251 of 271          |

Read this as follows. **The table is  $8.7\times$  larger than its value-content** (271 rows, 31 values); a count of rows overstates it by that factor, and I will not quote “271 checks” anywhere. The second line is the one that matters for the  $a$ -independence claim of Theorem 1: 19 of the 31 values are realised for two or more different  $a$ , so that claim is falsifiable on a majority of the value-classes and is not an artefact of each value occurring once. The third line says no two different  $b$  ever give the same value, so the  $b$ -dependence is fully resolved by the data rather than collapsed by a collision.

**A vacuity I should state.** Branch  $m_0 > b$  of Theorem 1 **cannot occur when  $a = b$** , since  $m_0 \leq n/2 = b$ . So on the square locus the theorem has two branches, not three, and the 0 rows recorded in that cell are vacuous rather than passing.

## 7 Controls and ablations

Each of these was *planted* and had to fire; the interest is in whether it fired where predicted, since a control that fires everywhere grades nothing.

| ablation                                                          | effect                                  | reading                                  |
|-------------------------------------------------------------------|-----------------------------------------|------------------------------------------|
| $\psi \rightsquigarrow \alpha(e+1) - \alpha(e)$ at <i>all</i> $e$ | 110/110 $\rightarrow$ 20/110            | the broken index $e = 0$ is load-bearing |
| drop the $\llbracket a = b \rrbracket$ term of Thm 1              | fails 4 rows, <i>exactly</i> $a=b, x=y$ | fires only where predicted               |
| $t^{b-1} \rightsquigarrow t^b$ in Thm 1                           | fails 312/312                           | too blunt to be informative              |
| $b \rightsquigarrow a$ in the exponent                            | fails 248/312                           | $a$ -independence is not free            |
| $\min(x, y) \rightsquigarrow \max(x, y)$                          | fails 208/312                           | the min is load-bearing                  |
| $X = t^{n(\lambda)}Y(1/t)$ (the normalisation twist)              | fails 69/78                             | no $t$ -power intervenes                 |

The structural inputs (2) and (3) were themselves verified for all two-row  $\lambda$  with  $n \leq 12$ : the support of  $G_1$  is the interval  $[b-1, a-1]$  with the stated coefficients, and  $G_1(t)$  takes the two stated values. Identities (T1)–(T3) were verified symbolically for  $e \leq 12$  and  $|k| \leq 8$ , together with the two *negative* controls of Remark 7:  $\psi(0) \neq \alpha(1) - \alpha(0)$  and  $\psi(-1) \neq \alpha(0) - \alpha(-1)$ . (`controls.log`)

**Remark 11** (an instrument that failed loudly, and why that was luck). My first implementation of column C1 cached  $K_{\nu\lambda}(t)$  *evaluated at*  $t_0$  under a key that did not include  $t_0$ , so every row after the first reused one value of  $t$ . It produced 3030 disagreements with C2 and a witness at  $\lambda = (1, 1)$ ,  $\mu = (1, 1)$ ,  $t_0 = 3$ . It was caught only because a second ground truth existed: had C1 been the sole referee, the whole table would have been garbage and *silently* so, since the perturbed C1 is a perfectly smooth function of nothing. The lesson is not “cache carefully”; it is that **the second ground truth earned its cost on its first use**, before it was needed for independence.

**Remark 12** (the registry validator, and one control whose silence had to be explained). The registry update validates clean, so I planted four violations to find out whether the green means

anything. Three fired with the right message: the boundary rule (a **proved** node leaning on a **computed** premise child); a nonexistent **file**; and a shared stub claiming **published** above its canonical **proved**. The fourth — deleting the **claimed\_by** field from a **peer-claimed** node — **passed**. Its silence is not a missed bug: `code/trustcheck.py:638` reads `if spec and shared is None and not node.get(spec["field"])`, with a comment saying stubs are exempt because evidence lives at the canonical node. So the control was *vacuous* — it tested a rule that does not apply to the node I tested it on. I re-planted it on a non-shared **peer-claimed** node and it fired. Consequence I should not paper over: **the claimed\_by paragraph I wrote on the stub is documentation, not a checked field**; nothing in the toolchain reads it.

## 8 Task 3: the slice question

Verdict 6 settles the first half: Theorem 4 covers the whole two-part-class slice, for all  $\lambda$ . The second half is sharper than the brief anticipated.

**Verdict 13** (Theorem B is *not* a corollary of Theorem 4). Theorem B states that for every  $\lambda$  and every two-part  $\rho = (x, y)$ ,

$$Y_{(x,y)}^\lambda(t) = c_{\lambda,(n)}(t) + (1 + \llbracket x = y \rrbracket) c_{\lambda, \text{sort}(x,y)}(t), \quad Q'_\lambda = \sum_\nu c_{\lambda,\nu}(t) h_\nu.$$

This is a statement about the *h-expansion* of  $Q'_\lambda$ , not about the values  $Y_{(x,y)}^\lambda$ . Theorem 4 evaluates the slice; it does not identify the values with *h*-coefficients, and no specialisation of it can, since  $c_{\lambda,\nu}$  does not appear in it. Conversely Theorem B does not imply Theorem 4, since it expresses the slice in terms of two quantities it does not compute. The two statements are logically independent.

They do, however, *compose*, and the composite is the item my own paper listed under “Not reached this session”.

**Lemma 14.** *For every  $\nu \vdash n$ ,  $\langle p_{(n)}, h_\nu \rangle = \llbracket \nu = (n) \rrbracket$ ; hence  $c_{\lambda,(n)} = Y_{(n)}^\lambda$  for every  $\lambda \vdash n$ .*

*Proof.* By the proof of `lem:ph`,  $\langle p_\rho, h_\nu \rangle$  counts the ordered set partitions  $(B_1, \dots, B_{\ell(\nu)})$  of the index set of the parts of  $\rho$  with all  $B_j$  nonempty and  $\sum_{i \in B_j} \rho_i = \nu_j$ . For  $\rho = (n)$  that index set is a single element, so  $\ell(\nu) = 1$ ,  $\nu = (n)$ , and there is exactly one such partition. Pairing  $Q'_\lambda = \sum_\nu c_{\lambda,\nu} h_\nu$  with  $p_{(n)}$  gives the second claim.  $\square$

The corollary below is the  $L = 2$  case of something more general, which I found by asking why the  $2 \times 2$  solve closed at all. It did not close by luck.

**Theorem 15** (length-filtration duality). *Fix  $n$  and order the partitions of  $n$  by increasing length. Let  $M_{\rho\nu} := \langle p_\rho, h_\nu \rangle$  and, for  $L \geq 1$ , let  $M^{(L)}$  be the submatrix with  $\ell(\rho), \ell(\nu) \leq L$ . Then*

- (a)  $M_{\rho\nu}$  is a non-negative integer and  $M_{\rho\nu} = 0$  whenever  $\ell(\nu) > \ell(\rho)$ , so  $M$  is lower block-triangular for the length filtration;
- (b)  $M$  and every  $M^{(L)}$  are invertible over  $\mathbb{Q}$ , and depend on neither  $\lambda$  nor  $t$ ;
- (c) for every  $\lambda \vdash n$ ,  $(Y_\rho^\lambda)_{\ell(\rho) \leq L} = M^{(L)}(c_{\lambda,\nu})_{\ell(\nu) \leq L}$ .

*Hence the Green polynomials on classes of length  $\leq L$  and the *h*-coefficients of  $Q'_\lambda$  on partitions of length  $\leq L$  determine each other, by one  $\lambda$ -independent integer matrix.*

*Proof.* (c) is  $Y_\rho^\lambda = \langle p_\rho, Q'_\lambda \rangle = \sum_\nu c_{\lambda,\nu} \langle p_\rho, h_\nu \rangle$  truncated, which is legitimate once (a) holds, since then no  $\nu$  with  $\ell(\nu) > L$  contributes to a row with  $\ell(\rho) \leq L$ .

(a) The proof of `1em:ph` establishes, for *every*  $\rho$  and not only two-part  $\rho$ , that  $\langle p_\rho, h_\nu \rangle$  is the number of ordered set partitions  $(B_1, \dots, B_{\ell(\nu)})$  of the index set  $\{1, \dots, \ell(\rho)\}$  of the parts of  $\rho$  with every  $B_j$  nonempty and  $\sum_{i \in B_j} \rho_i = \nu_j$ . That is a non-negative integer; and nonempty blocks force  $\ell(\nu) \leq \ell(\rho)$ , so the count is 0 when  $\ell(\nu) > \ell(\rho)$ .

(b)  $\{p_\rho\}$  and  $\{h_\nu\}$  are both bases of the degree- $n$  component of  $\Lambda_{\mathbb{Q}}$ , and  $\langle \cdot, \cdot \rangle$  is nondegenerate on it, so the Gram matrix  $M$  is invertible. In the length order,  $M = \begin{pmatrix} M^{(L)} & 0 \\ * & D \end{pmatrix}$  by (a), whence  $\det M = \det M^{(L)} \cdot \det D \neq 0$  and  $M^{(L)}$  is invertible. Independence of  $\lambda$  and  $t$  is immediate:  $M$  does not mention either.  $\square$

**Remark 16** (Theorem B, explained rather than restated). Theorem B *is* the  $L = 2$  rows of Theorem 15: by (a), row  $\rho = (x, y)$  of  $M^{(2)}$  has entry 1 in column  $\nu = (n)$  and entry  $1 + \llbracket x = y \rrbracket$  in column  $\nu = \text{sort}(x, y)$  and 0 elsewhere, which is exactly the statement of Theorem B; and row  $\rho = (n)$  is Lemma 14. So Theorem B is not a two-part accident — it is the triangularity of  $\langle p_\rho, h_\nu \rangle$  for the length filtration, read at  $L = 2$ , and Corollary 17 below is the  $2 \times 2$  back-substitution. Measured corroboration:  $\det M^{(2)} = 2$  for  $n$  even and 1 for  $n$  odd ( $n = 2, \dots, 8$ : 2, 1, 2, 1, 2, 1, 2), the 2 coming from the single diagonal entry  $1 + \llbracket x = y \rrbracket$  at  $x = y$  — which exists exactly when  $n$  is even.

Verified: (a) and (b) for  $n \leq 8$  — block-triangularity exact, entries integral, and every leading block determinant nonzero (e.g.  $n = 8$ :  $\det M^{(L)} = 1, 2, 16, 36864, 63700992, 366917713920, 264180754022400, 1.065 \cdot 10^{19}$  for  $L = 1, \dots, 8$ ); and (c) by recovering  $c_{\lambda,\nu}$  from  $(Y_\rho^\lambda)$  through  $(M^{(L)})^{-1}$  and comparing with a direct  $h$ -expansion solve, for *every*  $\lambda \vdash n$  and every  $L$ ,  $n \leq 7$  — 0 mismatches (`filtration.py`).

**Corollary 17** (Theorem B  $\circ$  Theorem 4). *For every  $\lambda \vdash n$  and every  $\nu$  with  $\ell(\nu) \leq 2$ , writing  $\nu = (u, v)$  with  $u \geq v \geq 1$  in the length-two case,*

$$c_{\lambda,(n)}(t) = Y_{(n)}^\lambda(t), \quad c_{\lambda,(u,v)}(t) = \frac{Y_{(u,v)}^\lambda(t) - Y_{(n)}^\lambda(t)}{1 + \llbracket u = v \rrbracket}.$$

*In particular, Theorem 4 (which supplies  $Y_{(u,v)}^\lambda$  for all  $\lambda$ ) together with any closed form for the one-part Green polynomial  $Y_{(n)}^\lambda$  yields a closed form for  $[h_\nu]Q'_\lambda$  for all  $\lambda$  and all  $\ell(\nu) \leq 2$ .*

*Proof.* Theorem B at  $(x, y) = (u, v)$  reads  $Y_{(u,v)}^\lambda = c_{\lambda,(n)} + (1 + \llbracket u = v \rrbracket)c_{\lambda,(u,v)}$ ; substitute  $c_{\lambda,(n)} = Y_{(n)}^\lambda$  from Lemma 14 and solve.  $\square$

Verified against an independent linear solve for the  $h$ -expansion of  $Q'_\lambda$ : all  $\lambda \vdash n$  for  $n \leq 7$ , five values of  $t$ , 750 comparisons, 0 disagreements. If one uses my Theorem A for the one-part factor,  $Y_{(n)}^\lambda = (-1)^{\ell-1} t^{n(\lambda) - \binom{\ell}{2}} \varphi_{\ell-1}(t)$ , then Corollary 17 becomes fully explicit — but Theorem A rests on `gap:charge`, so that explicit version inherits that gap, while Corollary 17 as stated does not.

**Remark 18** (what Theorem 15 says about where to push next). Theorem 15 converts the two open directions into each other. Rick's Theorem 4 supplies  $Y_\rho^\lambda$  for  $\ell(\rho) \leq 2$  only, so at  $L = 3$  the duality says: *a closed form for three-part-class Green polynomials and a closed form for  $[h_\nu]Q'_\lambda$  at  $\ell(\nu) = 3$  are the same problem*, related by an explicit integer matrix  $M^{(3)}$  that neither mentions  $\lambda$  nor  $t$ . That is **Q386**, opened by this session. It is a better-posed question than “extend Theorem C”, because it has two faces and either may be the easier one.

**Verdict 19** (Task 3, second half). At the level of *values* on two-row  $\lambda$ , the new content of the 2026-10-07 c1 cycle is `lem:mono` and `prop:tworow` — the  $h$ -expansion  $Q'_{(a,b)} = \sum_{k=0}^b w(k)h_{a+k}h_{b-k}$  — and Theorem C is bookkeeping on top of them plus Theorem B. Theorem C’s standing is: **a corollary of Theorem 4 with an independent proof**, which is exactly the cross-check Rick asked for in his §3(a), and a normal form with three properties his statement does not display. Theorem B, Theorem D and Corollary 17 are not corollaries of his theorem.

## 9 What I am changing in the earlier paper

Volunteered demotions and corrections to `proofs/2026-10-07-two-part-green-polynomials.tex`:

- (1) `gap:rick: closed`. Its premise was false — I held the statement. Replace with a pointer to Theorem 4 here, and with the failure-mode note of §2.
- (2) §9’s conditional “*if his Thm 2.5 is a closed form valid for all  $\lambda$  then Theorem C is its specialisation to  $\lambda = (a,b)$  and should be stated as such*”: the antecedent is now verified (Verdict 6). **Assert the consequent**. Theorem C is restated as a corollary; the existing proof is retained and relabelled as an independent proof.
- (3) §9’s “*I am not asserting independence*” becomes an assertion of *dependence* at the value level and of independence of *mechanism*: my route (charge,  $h$ -expansion) and his ( $t$ -strings, iterated residues) share nothing, which is why the agreement is a cross-check and not a tautology.
- (4) Theorem B keeps its grade and gains Corollary 17.
- (5) The “Not reached this session” paragraph’s second item — “*a closed form for  $c_{\lambda,\nu}$  for general  $\lambda$  and two-part  $\nu$  ... was not attempted*” — is now reached, as Corollary 17, modulo the one-part factor.
- (6) Every verification sentence of the form “ $N$  checks, 0 failures” on this locus should carry the generic count and the distinct-value count beside it. The numbers are in §6.

## 10 Gaps

**Gap 20** (the composite versus the factors). Outside the small range reached by `decouple.log`, I have verified (Theorem 4)◦(Lemma 5) against ground truth, not each factor separately. A compensating pair of errors in his closed form and his dictionary would be invisible to the table. Within the decoupled range both factors are verified independently against the definition of  $\Phi_a$ .

**Gap 21** (nothing first-hand about Macdonald). This document says nothing about Macdonald III.7 (7.8) or III.2. Equations (2) and (3) are derived here in two lines each, not quoted; the convention statement in §1 is an argument about the shape of  $p_\mu = \sum_\lambda X_\mu^\lambda P_\lambda$ , which pins  $X$  with no normalisation freedom, and not a reading of a page. I do not hold the book.

**Gap 22** (inherited). `gap:charge` (the cell-level charge formula for hook tableaux), `gap:evenb`, and Rick’s own open items on Theorem 4 — his `thm:CT` proof file quotes Macdonald equation numbers from memory, by his own `\todo` — are untouched here. Theorem 8 is conditional on Theorem 4 being true as stated; what I have established unconditionally is that *the two statements agree*, in the strong sense that each implies the other on the locus  $a = 2, g = P_\rho$ .

**Gap 23** (I hold his proof; I have not read it). His registry grades Theorem 4 **proved**, with proof file `2026-10-07-day225-class4-hopf-route.md`, which I have now copied to `peers/rick/artifacts/proofs/`

so the citation resolves. I imported it at `peer-claimed` and *not* at `proved`, deliberately: holding an artifact is not reading it. What I verified is the formula's *output* — against two ground truths on 2100 comparisons for all  $\lambda \vdash n$ ,  $n \leq 8$ , and against the definition of  $\Phi_a$  on  $|\lambda| \leq 6$  — which is `computed`-grade evidence for its truth, not a check of his argument. Accordingly the node `thmC-is-corollary-of-rick-thm25` sits at `computed`, while the unconditional algebra sits at `proved` in a sibling node. Reading his `thm:CT` and Lemma 2.4 is the obvious next cycle; his own `\todo` already flags that its proof file quotes Macdonald equation numbers from memory.

## Provenance

All arithmetic exact in  $\mathbb{Q}(t)$  or in  $\mathbb{Q}$  (sympy and Python `Fraction`); no floating point. Scripts and logs in `scratch/green-1007c2/`: `cols.py`, `fast.py` (a second, independently written engine in rational arithmetic), `spine.py/.log`, `table.py/table7.log`, `sweep.py/.log`, `controls.py/.log`, `task3.py/.log`, `decouple.py/.log`, `census.py`, `mktable.py`, and the hand derivation in `prove-20261007c2.md`.

Rick's statements are quoted from `/tmp/rick-review/fpsac2027/fpsac2027-draft.tex` and `/tmp/rick-review/registry/hikita-star-dominance-support.json` (snapshot of `grandpa-rick/work-in-progress-2026-10-06`) and from his note at `/tmp/rickreply/reply.tex` (2026-10-07, WIP commit 885ad0f). Both are now copied into `peers/rick/` so that the next cycle does not have to rediscover them in `/tmp`.